

Deciphering interventional dynamical causality from non-intervention complex systems

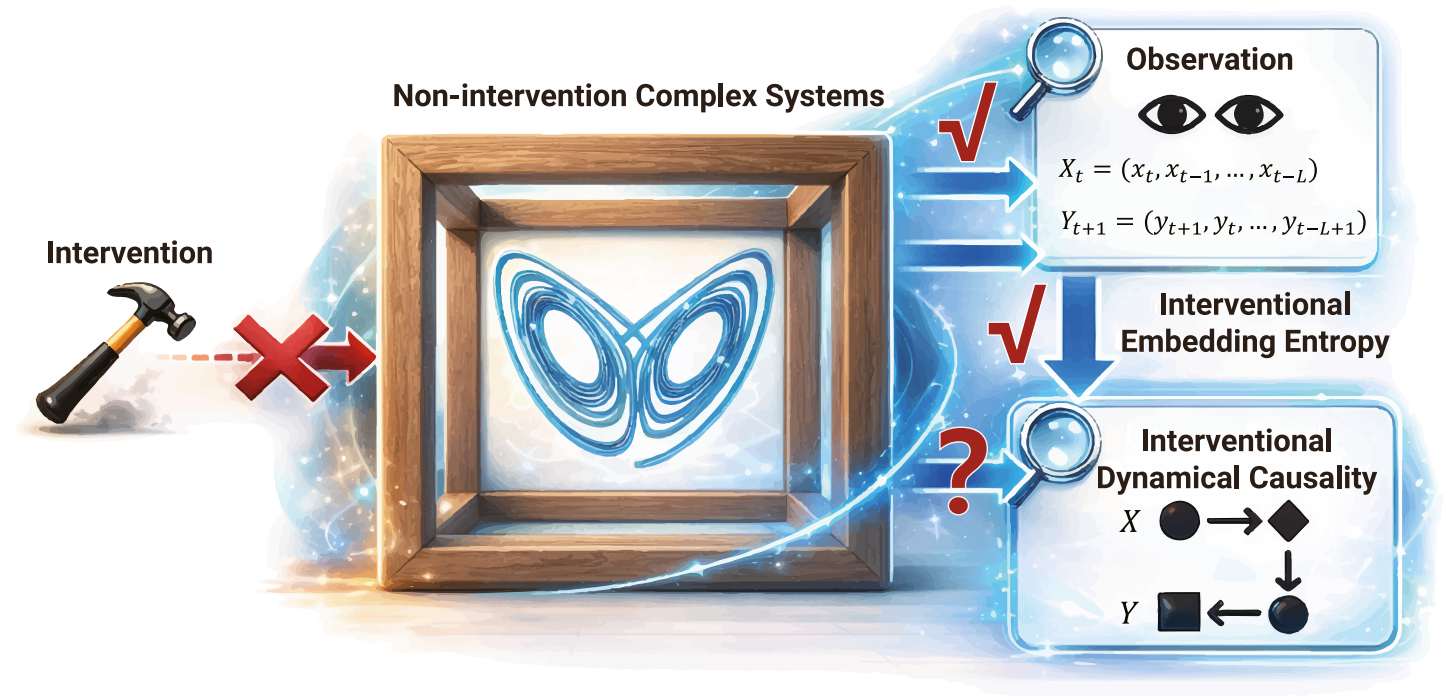
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GRAPHICAL ABSTRACT



PUBLIC SUMMARY

- A theoretical framework for interventional dynamical causality is proposed.
- Dynamical systems theory is integrated with modern causal inference.
- The interventional embedding entropy (IEE) method quantifies causal strength directly from observational data.
- The IEE approach enables robust, effective causal detection across diverse non-intervention complex systems.

Deciphering interventional dynamical causality from non-intervention complex systems

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Detecting and quantifying causal influence is a focal topic in the fields of science, engineering, and interdisciplinary studies. However, causal studies on non-intervention systems attract much attention but remain extremely challenging. The delay-embedding technique provides a promising approach. In this study, we propose a framework named interventional dynamical causality (IntDC) in contrast to the traditional constructive dynamical causality (ConDC). ConDC, including Granger causality, transfer entropy, and convergence of cross-mapping, measures the causality by constructing a dynamical model without considering interventions. A computational criterion, interventional embedding entropy (IEE), is proposed to measure causal strengths in an interventional manner. IEE is an intervened causal information flow but in the delay-embedding space. Further, the IEE theoretically and numerically enables the deciphering of IntDC solely from observational (non-interventional) time-series data, without requiring any knowledge of dynamical models or real interventions in the considered system. In particular, IEE can be applied to rank causal effects according to their importance and construct causal networks from data. We conducted numerical experiments on logistic dynamics, coupled-Hénon maps, and chaotic neural networks to demonstrate that IEE can find causal edges accurately, eliminate effects of confounding, and quantify causal strength robustly over traditional indices. We also applied IEE to real-world tasks, including estimating neural connectomes of *Caenorhabditis elegans*, detecting COVID-19 transmission networks in Japan, and investigating regulatory networks surrounding key circadian genes. IEE performed as an accurate and robust tool for causal analyses solely from the observational data. Both the IntDC framework and the IEE algorithm proposed in this study provide an approach to understanding causal influence in diverse non-intervention complex systems.

INTRODUCTION

Directional or bidirectional interactions among different components give rise to a variety of complex phenomena in nature and social society. The causality or the causal effect is one of the most attractive directional relations originating from the non-reversibility of time. Modeling, detecting, and quantifying causal influence from observational data are crucial for describing, interpreting, predicting, and even controlling complex systems.

Statisticians believe that causality is contained and obtainable in universal random variables regardless of the time label. Neyman-Rubin's potential outcome framework,^{1–3} the Wright's structural equation model,^{4,5} and the Pearl's causal diagram model^{6,7} are the most famous statistical approaches and have been proved to be of mathematical equivalence.⁸ Especially, the instrumental variable method is widely applied in the study of causality among economics, environments, and other disciplines.⁹ Statistical methods seek to discern binary causal relations among random variables on a directed acyclic graph without relying on temporal data. However, in general complex dynamical systems, the causation must precede the effect, feedback is common, and the

causal strength needs to be quantitatively measured. Time should definitely play a key role.

Dynamics-oriented researchers have proposed fruitful algorithms for measuring the observational causality from time series, commonly referred to as the dynamical causality.¹⁰ The celebrated Granger causality (GC)^{11,12} employs a linear model and uses improvement of predictability over time to illustrate the causality. Transfer entropy (TE)¹³ generalizes GC to the nonlinear case by quantifying the prediction uncertainty through Shannon entropy. GC and TE rely on the separability assumption, namely that removing one variable from the system does not influence the others. Neither GC nor TE addresses the “non-separability” problem, in which removing a causal variable inevitably alters the dynamics of downstream variables. A systematic discussion of the separability issue can be found in recent studies.^{10,14–16} To measure causality in non-separable systems, numerous approaches have emerged in the last decade within the framework of delay embedding. These include convergence of cross-mapping (CCM),^{14,17} partial cross-mapping (PCM),¹⁵ continuity scaling,¹⁶ inverse continuity,¹⁸ topological expansion,¹⁹ joint distance distribution,²⁰ embedding entropy (EE)¹⁰ and other indices.^{21,22} Besides, Runge et al. designed the PCMC method to further remove “common drivers” by selecting parent nodes iteratively.²³ Friston et al. proposed the dynamical causal modeling (DCM), which is a Bayesian fitting from data with pre-selected models.²⁴ DCM was originally developed for modeling neural dynamics.²⁵ Krakovská et al. conducted a comparative analysis of six methods for detecting causality in bivariate systems.²⁶

Nonetheless, most indices designed for quantifying causality from time series primarily focus on estimating directional causal relationships at the constructive level, referred to as constructive dynamical causality (ConDC). Traditionally, to detect and quantify causality at the interventional level, which is termed interventional dynamical causality (IntDC) in this study, requires intelligent modulation and manipulation of the dynamical system. By allowing external intervention to the system and recording data under different perturbations, frameworks such as perturbation cascade inference (PCI)²⁷ and dynamical causal effect (DCE)^{28,29} provide relevant computational schemes. To detect asymmetry information transfer in known two-dimensional dynamical systems, Liang and Kleeman proposed an analytic approach named Liang-Kleeman information flow,^{30–32} which serves as a prototype for IntDC. This method quantifies causality by freezing one variable as a parameter and evaluating the resulting outcomes using the Frobenius-Perron operator. The Liang information flow has been further extended to multivariate, stochastic, and quantum systems.^{33–36} However, due to ethical or practical limitations, many real systems should be analyzed without any external intervention. An essential problem is how to measure IntDC solely from the observational data but at the interventional level.

In this study, we propose the IntDC framework and introduce a criterion named interventional embedding entropy (IEE), which aims to identify and quantify IntDC between variables solely from the observational time series. Actually, IEE is rigorously derived by the theory of the delay embedding. Numerical IEE does not require specific prior knowledge of dynamics, and additional perturbation to the system is also unnecessary. Compared to ConDC indices, such as

GC, TE, and CCM, IEE designed for IntDC has the capability to rank the importance of causal effects and construct directional causal networks more effectively. We demonstrate numerical experiments on both simulated examples and real datasets, including estimating neural connectomes of *Caenorhabditis elegans* (*C. elegans*), evaluating COVID-19 transmission in Japan, and constructing regulatory networks surrounding key circadian genes.

MATERIALS AND METHODS

ConDC

The definition of ConDC for a complex system in the original time-series space is given as follows:

Definition 1 (Constructive dynamical causality, ConDC¹⁰). For a complex dynamical system,

$$\mathbf{x}_{t+1} = \mathbf{f}(\mathbf{x}_t, \mathbf{x}_{t-1}, \dots, \mathbf{x}_{t-p}) + \mathbf{e}_t, \quad (\text{Equation 1})$$

where $\mathbf{x} = (x^{(1)}, x^{(2)}, \dots, x^{(n)})^T$ represents the system with n components, $\mathbf{f} = (f_1, f_2, \dots, f_n)^T$ is a vector function, p is referred to the memory time, and $\mathbf{e}_t$ is an independent noise term, there exists ConDC from component $x^{(j)}$ to $x^{(i)}$, if $\exists k \in \{0, 1, 2, \dots, p\}$ such that $\partial f_i / \partial x_{t-k}^{(j)} \neq 0$ for almost any t .

For simplicity, we consider a discrete two-variable dynamical system as an example:

$$\begin{cases} x_{t+1} = g(x_t, x_{t-1}, \dots, x_{t-p}, e_{x,t}), \\ y_{t+1} = f(x_t, x_{t-1}, \dots, x_{t-p}, y_t, y_{t-1}, \dots, y_{t-p}, e_{y,t}), \end{cases} \quad (\text{Equation 2})$$

where x, y are two variables, p denotes the time step during which causality is considered, and $e_{x,t}$ stand for small noise terms. According to Definition 1, there exists ConDC from x to y since the evolving equation of y_{t+1} depends on the historical behavior of x , while there is no ConDC from y to x as the dynamics of x_{t+1} is independent of y . GC and TE measure the predictability of $\mathbf{f}$ in the original time-series space to quantify the ConDC from x to y through linear regression and entropy uncertainty, respectively (Figure 1B). The causal strength of GC/TE is based on the assumption of separability, which means that removing x from the system does not influence the observed data of y . However, in general coupled systems, the separability is not satisfied, as removing one variable changes the values of affected ones.^{14–16}

To address the challenge of detecting ConDC in universal non-separable systems, various approaches from the delay-embedding space have been proposed. In the delay-embedding framework, the bivariate system Equation 2 is assumed to evolve into an attractive manifold with an inner dimension d . Let the time-delayed vectors of x and y be

$$\mathbf{X}_t = (x_t, x_{t-1}, \dots, x_{t-L})^T \in \mathcal{M}_X \subseteq \mathbb{R}^{L+1}, \quad (\text{Equation 3})$$

and

$$\mathbf{Y}_{t+1} = (y_{t+1}, y_t, y_{t-1}, \dots, y_{t-L+1})^T \in \mathcal{M}_Y \subseteq \mathbb{R}^{L+1}, \quad (\text{Equation 4})$$

where L is the time-delayed length, and $\mathcal{M}_X$ and $\mathcal{M}_Y$ represent the manifolds formed by $\mathbf{X}_t$ and $\mathbf{Y}_t$, respectively. According to the seminal stochastic version of Takens' embedding theorem,^{37–40} we can obtain the following theorem:

Theorem 1 (ConDC in delay-embedding space). If x is the ConDC of y in dynamics Equation 2, and $\mathbf{X}_t, \mathbf{Y}_t$ are the time-delayed vectors, respectively, then there exists a smooth projection operator $\mathbf{F}$ in generic sense such that

$$\mathbf{X}_t = \mathbf{F}(\mathbf{Y}_{t+1}), \quad (\text{Equation 5})$$

when the time-delayed length satisfies $L \geq 2d$, where d is the inner dimension of the attractive manifold.

The detailed derivation can be referred to the supplemental information. The "generic sense" means that a smooth projection exists for a dense and open set of all possible time-delayed ways.^{37,38} Equation 5 shows that the causal variable $\mathbf{X}_t \in \mathcal{M}_X$ can be reconstructed by the effect variable $\mathbf{Y}_{t+1} \in \mathcal{M}_Y$. However, as $\mathbf{F}$ is not reversible in generic sense, we cannot determine $\mathbf{Y}_{t+1}$ only by the information from $\mathbf{X}_t$. According to Theorem 1, the causal dependence in $\mathbf{f}$ between x and y is transformed into the reconstructability of $\mathbf{F}$ (Figure 1D). Theorem 1, established for the dynamical system in Equation 2, can be generalized to multi-variable systems as described in Equation 1, allowing for feedback interactions between different variables. Instead of fitting a model by removing the causal variable as in GC and TE, detecting the existence and quantifying the continuity characteristics of $\mathbf{F}$ in the delay-embedding space can be adequate for causal identification, especially for universal non-separable systems. Related algorithms include CCM, PCM, and IEE.

Figures 1A, 1B, and 1D summarize dynamical causality at the constructive level, where the ConDC from variable x to y is estimated only from the historical observed time series. However, fitting Equation 2 or Equation 5 from data is criticized as mere association or prediction.^{41,42}

IntDC

The detection of hidden essential-level causality necessitates intervention or manipulation to the system.^{8,27,43,44} In this study, we propose the definition of IntDC for a complex system in the original time-series space as the following.

Definition 2 (Interventional dynamical causality, IntDC). For a complex dynamical system Equation 1, one component $x^{(j)}$ is the IntDC of another component $x^{(i)}$, if $\exists k \in \{0, 1, 2, \dots, p\}$ such that $\delta x_{t+1}^{(i)} = \tilde{x}_{t+1}^{(i)} - x_{t+1}^{(i)}$ depends on $\delta x_{t-k}^{(j)} = \tilde{x}_{t-k}^{(j)} - x_{t-k}^{(j)}$ for almost any t , where $\tilde{x}_{t-k}^{(j)}$ is the intervened dynamics of $x^{(j)}$, and $\tilde{x}_{t+1}^{(i)}$ is the dynamics of affected $x^{(i)}$ after the intervention on $x^{(j)}$.

The intervention $\delta x_{t-k}^{(j)}$ can be either a pulse stimulation or persistent disturbances to the system. Figure 1C illustrates the IntDC in the time-series space for the simplified bivariate system Equation 2, involving variables x and y . To measure the IntDC, data before and after the intervention, i.e., (x_t, y_t) and $(\tilde{x}_t, \tilde{y}_t)$, are typically required, as in

PCI and DCE. PCI measures the distances of every node in the network from each perturbed node to reconstruct the directed causal network.²⁷ DCE classified causal coupling into four situations based on the type of effects (stationary statistic change, phase orbit change) and the type of interventions (state space interventions, parametric interventions).²⁸ These conventional algorithms rely on reproducible dynamics or data availability under different settings.

To address challenges of causal inference in widespread non-intervention, nonlinear, and non-separable systems, we turn to the study of intervened dynamics of Equation 5 in the delay-embedding space. Denote $\delta \mathbf{X}_t = \tilde{\mathbf{X}}_t - \mathbf{X}_t$ and $\delta \mathbf{Y}_{t+1} = \tilde{\mathbf{Y}}_{t+1} - \mathbf{Y}_{t+1}$, where $\tilde{\mathbf{X}}_t$ and $\tilde{\mathbf{Y}}_{t+1}$ are the time-delayed vectors of x and y after the intervention, respectively. We can obtain the following theorem:

Theorem 2 (IntDC in delay-embedding space). If x is the IntDC of y in dynamics Equation 2, and $\mathbf{X}_t, \mathbf{Y}_t$ are the time-delayed vectors, respectively, then there exists a smooth projection operator $\mathbf{F}$ in generic sense such that

$$\delta \mathbf{X}_t = \nabla_{\mathbf{Y}} \mathbf{F}(\mathbf{Y}_{t+1}) \cdot \delta \mathbf{Y}_{t+1}, \quad (\text{Equation 6})$$

when the time-delayed length satisfies $L \geq 2d$ and the intervention is sufficiently small, where d is the inner dimension of the attractive manifold.

Theorem 2 is directly deduced by Theorem 1. However, Equation 6 provides insights on the causality under intervention. It indicates that $\delta \mathbf{Y}_{t+1}$ contains complete information of $\delta \mathbf{X}_t$ in the neighborhood of $\mathbf{Y}_{t+1}$ and can reconstruct $\delta \mathbf{X}_t$ if there exists IntDC from the causal variable x to the effect variable y (Figure 1E). The conclusion in Equation 6 of Theorem 2, derived from the dynamics in Equation 2, remains valid for any pair of components exhibiting IntDC in the multi-variable system of Equation 1, and feedback between different components is permitted. Figures 1A, 1C and 1E give an overview of the IntDC framework.

Regarding the relationship between Definition 1 and Definition 2, we claim that, in theory, the existence of IntDC between two variables is a sufficient condition for the existence of ConDC. IntDC can be interpreted as a dependence captured by the first-order differential form of ConDC in Equation 1. However, in practical situations, direct numerical estimation of ConDC suffers from false positives, particularly in the presence of confounders (discussed in the results and discussions).

IEE

We introduce the IEE criterion, specifically designed for quantifying IntDC solely from the observational data in the delay-embedding space.

Definition 3 (Interventional embedding entropy, IEE).

The IEE criterion is proposed to measure the average information retention or reconstructability from the effect $\delta \mathbf{Y}_{t+1}$ to the causation $\delta \mathbf{X}_t$, formulated as

$$\text{IEE}[x \rightarrow y] := \text{CMI}(\delta \mathbf{X}_t, \delta \mathbf{Y}_{t+1} | \mathbf{Y}_{t+1}). \quad (\text{Equation 7})$$

In Equation 7, CMI is the conditional mutual information,⁴⁵ i.e.,

$$\text{CMI}(\mathbf{x}, \mathbf{y} | \mathbf{z}) = \iiint p(\mathbf{x}, \mathbf{y}, \mathbf{z}) \log \frac{p(\mathbf{x}, \mathbf{y} | \mathbf{z})}{p(\mathbf{x} | \mathbf{z}) p(\mathbf{y} | \mathbf{z})} d\mathbf{x} d\mathbf{y} d\mathbf{z}. \quad (\text{Equation 8})$$

IEE provides a quantitative measure of causal strength that is comparable across different edges within the same dynamical system. When dealing with data, the $\delta \mathbf{Y}_{t+1}$ is

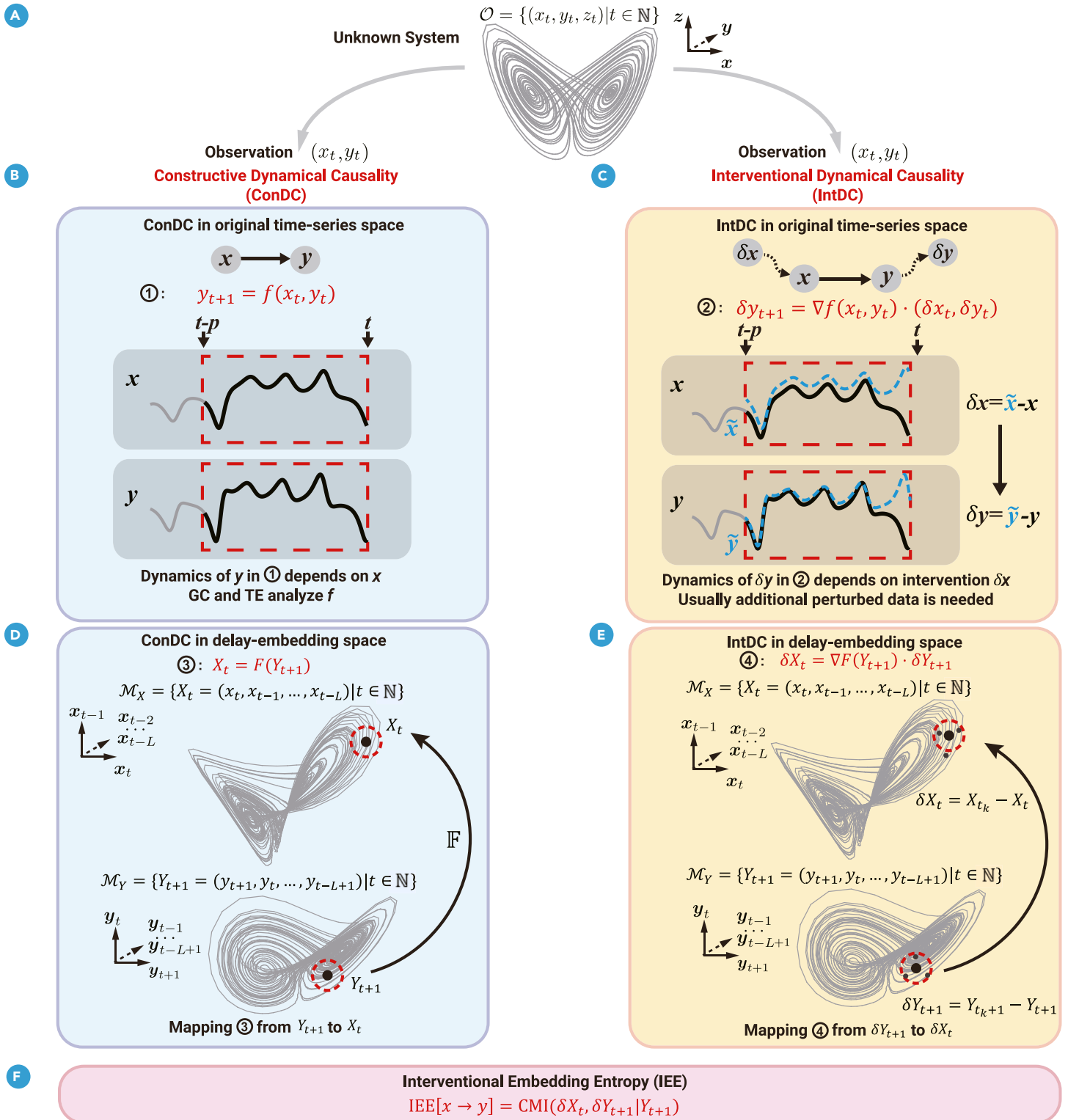


Figure 1. Illustration of the ConDC and IntDC (A) An illustrative example of a complex system where (x_t, y_t) are observed time series. The z_t stands for the other variable that is not the primary focus of analysis. (B) ConDC from x to y indicates that the dynamics of y depends on x . Algorithms such as Granger causality (GC) and transfer entropy (TE) infer the ConDC in the original time-series space. (C) IntDC from x to y indicates that the deviation of y , i.e., δy , depends on an intervention on x , i.e., δx . Usually additional data from a perturbed system $(\tilde{x}_t, \tilde{y}_t)$ are necessary for detecting IntDC in the original time-series space. The parameter p is the time lag after intervention occurs. (D) ConDC can be detected in the embedding space by cross-mapping F . The causal vector X_t can be reconstructed by the effect vector Y_{t+1} . Parameter L is time delay used in the delay embedding. (E) IntDC can be modeled in the embedding space. The interventional causal vector δX_t should be reconstructable by the effect vector δY_{t+1} around Y_{t+1} . In numerical computation, δY_{t+1} is approximated by the distance from Y_{t+1} to its neighbors Y_{t+1} , while δX_t is determined using X_{t_k} corresponding to the same time index t_k . (F) Interventional embedding entropy (IEE) measures the average information retention contained in δY_{t+1} from δX_t , quantifying the IntDC from variable x to y . Notably, IEE does not require additional perturbations to the system and can be inferred from observational data solely.

estimated by the distance from Y_{t+1} to its neighbored points Y_{t_k+1} in the embedding space, and δX_t is calculated by X_{t_k} with the same time label t_k (Figures 1E and 1F). Thus, Equation 7 can be numerically approximated only from the observational data, especially for non-inter-

vention systems. Table 1 presents the numerical algorithm for computing IEE, whose detailed descriptions and computational procedures are provided in the supplemental information.

Table 1. Algorithm: IEE

1. Given observed time series $\{x_{t_n}|t_n \geq 0\}, \{y_{t_n}|t_n \geq 0\}$.
2. Construct time-delayed vectors X_{t_n} and Y_{t_n+1} by Equations 3 and 4.
3. Get K nearest neighbors Y_{t_k+1} around Y_{t_n+1} , $k = 1, 2, \dots, K$.
4. Sample δY_{t_n+1} condition on Y_{t_n+1} as $Y_{t_k+1} - Y_{t_n}$.
5. Sample δX_{t_n} condition on Y_{t_n+1} as $X_{t_k} - X_{t_n}$ with the same time labels t_k .
6. Calculate Equation 7 around Y_{t_n+1} and average over n samples to approximate $IEE[x \rightarrow y]$.

Connections with statistical causal models and assumptions of ConDC/IntDC

The concept of dynamical causality in Equation 1 develops from traditional structural causal models, which typically studies structure equations of the form $Y = f(X, \epsilon, \theta)$, where X and Y are random variables, f is a function parameterized by θ , and ϵ represents noise or residue terms. The framework of ConDC has been systematically studied with three key generalizations beyond classical structural equation modeling¹⁰: (i) temporal information is fully considered by dynamical systems, ensuring that causes precede effects in time; (ii) feedback or causal loops between variables are permitted under the consideration of time delays, relaxing the restriction of directed acyclic graphs (DAGs); (iii) quantitative measures of causal strength are provided beyond qualitative identification, enabling direct comparisons of causality between different variables. The IntDC proposed in the study further integrates the concept of interventions and potential outcomes within the dynamical framework.⁹ IntDC aims to quantify causal strengths at the interventional level directly from observational data (without the need for additional interventions to the system, which are often impractical or infeasible in real-world datasets).

The following assumptions (A1–A8) clarify key connections between dynamical causality and statistical causal inference:

(A1) Temporal order assumption: causal information is embedded in time-series data and causes must precede their effects;

(A2) Causal sufficiency (unconfoundedness) assumption: all common drivers (confounders) are observed such that the causal graph accurately represents relationships among the observed variables;

(A3) Causal Markov assumption: each variable x_i is independent of its nondescendants given its direct causes (parents);

(A4) Faithfulness assumption: the causal graph structure can represent the conditional independence contained in the joint probability density;

(A5) Stationarity assumption: the dynamical system evolves into a stable attractive manifold, allowing stable causal relations to be measured from observational data;

(A6) Intervention ignorability assumption: given the system dynamics Equation 1, initial conditions x_0 and noise ϵ_t , the intervention on variable $x^{(j)}$ at time $t-k$ is independent of the potential outcome of $x^{(j)}$ at time $t+1$, i.e., $x_{t+1}^{(j)} \perp\!\!\!\perp \delta x_{t-k}^{(j)} | (f, x_0, \epsilon_t)$ for $k = 0, 1, 2, \dots, p$;

(A7) Stable infinitesimal intervention assumption: the average information retention from $x^{(j)}$ to $x^{(j)}$ in dynamics Equation 1 remains stable under sufficiently small interventions $\delta x^{(j)}$;

(A8) Consistency assumption: the outcome $\tilde{x}_{t+1}^{(j)}$ in Equation 1 can be precisely determined under known dynamics (f, x_0, ϵ_t) and a fixed intervention $\delta x_{t-k}^{(j)}$.

In practice, (A2) can be relaxed if the primary goal is to quantify the total causal strength (including both direct and indirect causality) between observed variables. In the results and discussions section, we demonstrate that the IEE criterion is capable of mitigating the impact of confounding variables, thereby providing robust causal estimates even when some confounders are unobserved. Some of these assumptions for causal discovery from observational time-series data have also been discussed by Runge et al.^{23,46}

GC, TE, and CCM are all criteria for ConDC, which estimate causal strengths by fitting constructive dynamical models (i.e., Equation 2 in the time-series space or Equation 5 in the delay-embedding space). GC employs linear vector regression to fit f in Equation 2. TE extends GC to nonlinear cases by quantifying information transfer between variables. CCM constructs a local-linear cross-mapping F in Equation 5 by the time-delayed embedding to capture causality. For a detailed comparison of these ConDC methods, readers are referred to Table 1 in Shi et al.¹⁰ IEE captures causality at the interventional level, whereas GC, TE, and CCM are restricted to association level causality. Moreover, IEE as one measure for IntDC can effectively avoid confounding effects in structures such as $x \leftarrow z \rightarrow y$ (discussed in the results and discussions).

RESULTS AND DISCUSSIONS

IEE captures the numerical behavior of causality in logistic systems robustly

To assess the numerical performance of measuring IntDC, we first applied the IEE algorithm on the following two-node logistic dynamics:

$$\begin{cases} x_{t+1} = 3.7[(1 - \beta_{yx})x_t(1 - x_t) + \beta_{yx}y_t(1 - y_t)] + \epsilon_{x,t}, \\ y_{t+1} = 3.7y_t[1 - (1 - \beta_{xy})y_t - \beta_{xy}x_t] + \epsilon_{y,t}, \end{cases} \quad (\text{Equation 9})$$

where $\epsilon_{\cdot,t}$ represents independent Gaussian noises, and parameters β_{xy} and β_{yx} modulate the IntDC strength from x to y and from y to x , respectively (see further details in supplemental information).

IEE can capture the numerical behavior of IntDC in the two-node logistic system. When we set $\beta_{xy} \equiv 0$ and let β_{yx} increase from 0 to 0.3, we observed a monotonic increase in $IEE[y \rightarrow x]$ (the red line with squares in Figure 2A), while $IEE[x \rightarrow y]$ remained around zero (the blue line with dots in Figure 2A). When we set $\beta_{xy} \equiv 0.1$ and let β_{yx} increase from 0 to 0.3, $IEE[y \rightarrow x]$ exhibited a monotonic increase (the red line with squares in Figure 2B), while $IEE[x \rightarrow y]$ stayed above zero (the blue line with dots in Figure 2B). The gray dashed lines in Figures 2A and 2B represent 0.01 for reference. A detailed comparison with GC, TE, and CCM is presented in Figures S1 and S2. The mean values and standard deviations of the indices over 100 simulation runs are provided in Tables S1–S4. Figures S1B and S2B indicate that GC exhibited non-monotonic behavior. TE $[y \rightarrow x]$ demonstrated a decreasing trend when β_{yx} increased, specifically for $\beta_{yx} \in (0, 0.02]$ in Figure S1C and Table S1 and $\beta_{yx} \in (0, 0.03]$ in Figure S2C and Table S3, which may lead to false-negative issues (ignorance of existence causality). CCM $[x \rightarrow y]$ remained significantly different from zero even when $\beta_{xy} = 0$ (in the range $\beta_{yx} \in [0.15, 0.3]$ in Figure S1D and Table S2). CCM $[y \rightarrow x]$ was also significantly different from zero when $\beta_{yx} = 0$ (with $\beta_{xy} = 0.1$ in Figure S2D and Table S3). These results suggest that CCM may suffer from the false-positive problem (incorrect identification of non-existence causality).

Within suitable parameter ranges, IEE demonstrates stability and robustness. We validated the robustness of the IEE algorithm under various conditions, including different delayed lags L , the numbers of nearest neighbors K , lengths of time series N , and noise standard deviations σ . The causal strengths measured by IEE exhibited a consistent descending trend as β_{yx} decreased from 0.15 to 0.05 through 0.125, 0.1, 0.075, as expected (Figures 2D–2G). These results support that IEE reliably preserves the correct ranking of causal strengths across different parameter settings with relatively stable variance.

IEE eliminates the influence of confounding variables

IEE remains applicable and accurate even in presence of confounding variables. We simulated a three-node logistic dynamics (x_t, y_t, z_t) , where z acted as a confounder (see details in supplemental information). There were consistently non-zero causal effects from z to x and from z to y , while β_{xy} adjusted the causal strength from x to y (Figure 2C). When $\beta_{xy} = 0$, $IEE[x \rightarrow y]$ was almost zero and significantly different from $IEE[z \rightarrow x]$ and $IEE[z \rightarrow y]$ (Figure 2C); but conventional ConDC indices such as GC/TE/CCM produced false-positive causality from x to y due to the presence of confounder z (Figure S3). According to the IntDC framework, an intervention on x (i.e., δx) will not induce an indirect change in y (i.e., δy) through z ; in other words, there is no intervention-induced pathway through $x \leftarrow z \rightarrow y$. However, traditional ConDC indices by fitting dynamical models may falsely infer an association between x and y due to the influence of the confounder z . IEE, specially designed for measuring IntDC, remained unaffected by the confounder z and yielded accurate results.

IEE enables causal network reconstruction with quantification of causal strengths

IEE has the capability to reconstruct the causal network structure and rank the causal influence between nodes in complex networks at the interventional level. We used a 10-node coupled-Hénon maps as an example, where each node serves as the dynamical cause to its subsequent node (Figure 3A). Further details regarding the dynamics can be found in supplemental information.

To verify the effectiveness of IEE in reconstructing causal networks, we calculated the area under the curve (AUC) values based on multiple simulated time series. The AUC value (mean $\pm$ standard deviation) of IEE was $0.871 \pm$

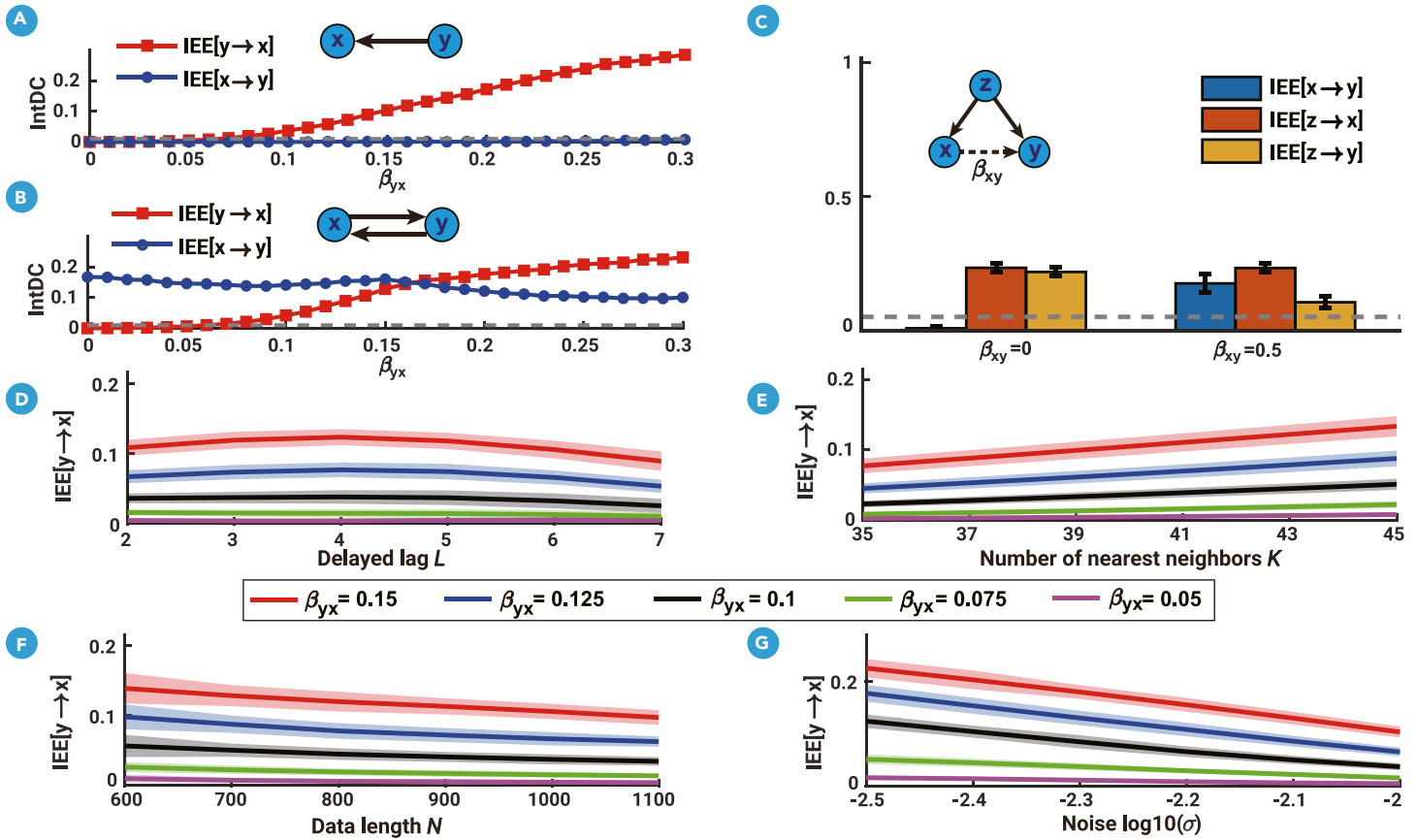


Figure 2. Performance of IEE on logistic systems (A and B) Using the two-node logistic system. The β_{xy} and β_{yx} control the coupling coefficients from x to y and from y to x , respectively. (A) The mean values of IEE when $\beta_{xy} = 0$ and β_{yx} ranges from 0 to 0.3 over 100 simulations. IEE[$y \rightarrow x$] (the red line with squares) increases monotonically, while IEE[$x \rightarrow y$] (the blue line with dots) stays around zero. (B) The mean values of IEE when $\beta_{xy} = 0.1$ and β_{yx} ranges from 0 to 0.3 over 100 simulations. IEE[$x \rightarrow y$] (the blue line with dots) is significantly positive. The gray dashed lines in (A) and (B) represent the constant 0.01 as reference. (C) The performance of IEE in the three-node logistic system, where z acts as a confounding variable. When $\beta_{xy} = 0$, IEE[$x \rightarrow y$] is almost zero correctly. IEE designed for measuring IntDC is not affected by confounders, which usually leads to false positives in ConDC. The results for $\beta_{xy} = 0.5$ are shown for a comparison. The gray dashed line is 0.05 for reference. (D–G) The robustness of IEE under different delayed lags L , the numbers of nearest neighbors K , data lengths N , and noise standard deviations σ in the two-node logistic system, respectively. The parameter $\beta_{xy} \equiv 0$, and β_{yx} takes values of 0.15 (red), 0.125 (blue), 0.1 (black), 0.075 (green), and 0.05 (magenta). Lines denote mean values under 50 simulations, and shaded areas represent the standard deviation. IEE consistently preserves the correct ranking of IntDC across various settings, with relatively stable variance.

0.008, significantly higher than conventional ConDC indices, such as GC with 0.707 ± 0.069 , TE with 0.816 ± 0.018 , and CCM with 0.837 ± 0.012 (boxplots in Figure 3B). The AUCs of IEE exhibited both a higher mean and a lower variance relative to other approaches. Additionally, receiver operating characteristic (ROC) curves for the four indices are presented in Figure S4.

To show the capability for ranking the importance of causal influences, we conducted two tests. We first calculated the IEE from node 1 to the other nine nodes. Consistent with the actual scenario, IEE displayed sequential decrease in IntDC originating from node 1 across nodes 2–10 (Figure 3C). Then, we measured the IEE received by node 7 from the other nine nodes. As expected, nodes 1–6 exhibited ascending causal strengths on node 7, whereas nodes 8–10 had little influence on node 7 (Figure 3D). IEE accurately discerned the IntDCs, with node 6 showing the strongest value. A comparison showed that GC and TE failed to rank the influence from node 1 accurately in the first test (Figure S5), while CCM suffered from the false-positive causal detection from node 8 to node 7 in the second test (Figure S6).

IEE quantifies IntDC without requiring additional experimental interventions

IEE is specifically designed for calculating IntDC solely from observed time-series data without requiring additional perturbations to the system. To validate its accuracy, we conducted a comparison against true perturbed deviations using chaotic neural networks (chNNs) as a model system.^{47,48}

The chNN comprises an output variable $\mathbf{x} = (x_1, x_2, \dots, x_{N_{\text{node}}})^T$ and two internal variables, with N_{node} representing the number of neurons in the network. We chose $N_{\text{node}} = 10$ in simulations. Details on the dynamics can be found in the supplemental information. We denoted the observed time-series data from the

stationary system without intervention as $\mathbf{x}_{\text{obs}}$. When the neuron i was removed (set as a constant zero), the perturbed data were denoted as $\mathbf{x}_{\text{per}}^{(i)}$. Removing one node was taken as an intervention to the system. The stationary probability density functions of the j th neuron before and after the intervention were represented by $p_j(\mathbf{x}_{\text{obs}})$ and $p_j(\mathbf{x}_{\text{per}}^{(i)})$, respectively. We used the Kullback-Leibler divergence (KLD), i.e.,

$$D_{ij} \triangleq \text{KLD}[p_j(\mathbf{x}_{\text{obs}}) \| p_j(\mathbf{x}_{\text{per}}^{(i)})] = \iint p_j(\mathbf{x}_{\text{obs}}) \log \frac{p_j(\mathbf{x}_{\text{obs}})}{p_j(\mathbf{x}_{\text{per}}^{(i)})} d\mathbf{x}_{\text{obs}} d\mathbf{x}_{\text{per}}^{(i)}, \quad (\text{Equation 10})$$

to quantify the true influence of the intervention, i.e., ground truth of the IntDC, from i to j . By conducting 100 randomly simulated chNNs with interventions (in each simulation, one node is removed from each chNN with N_{node} nodes), we recorded 1,000 KLD values and compared them with the results of IEE (calculated from $\mathbf{x}_{\text{obs}}$ solely). We observed that IEE[$x_i \rightarrow x_j$] exhibited a positive linear correlation with the KLD D_{ij} (with $R^2 = 0.869$) in Figure 4A, where gray dots are the 1,000 samples of causal edges between neurons and the red line stands for the linear regression. Results for GC (with $R^2 = 0.572$), TE (with $R^2 = 0.912$), and CCM (with $R^2 = 0.622$) indicated that GC and CCM could not linearly reflect KLD (Figures 4B–4D). Furthermore, we drew the violin plots of cosine similarity S_c (see supplemental information) between KLD and the four causal indices on 100 chNNs (Figure 4E). IEE (0.904 ± 0.024) had the highest mean similarity with KLD, compared to GC (0.711 ± 0.076), TE (0.888 ± 0.022), and CCM (0.687 ± 0.033). In addition, we calculated the rank-based correlation coefficient Kendall's tau. CCM ($\rho_{\tau}^{\text{CCM}} = 0.457$) showed the strongest rank-based correlation with KLD, compared with IEE

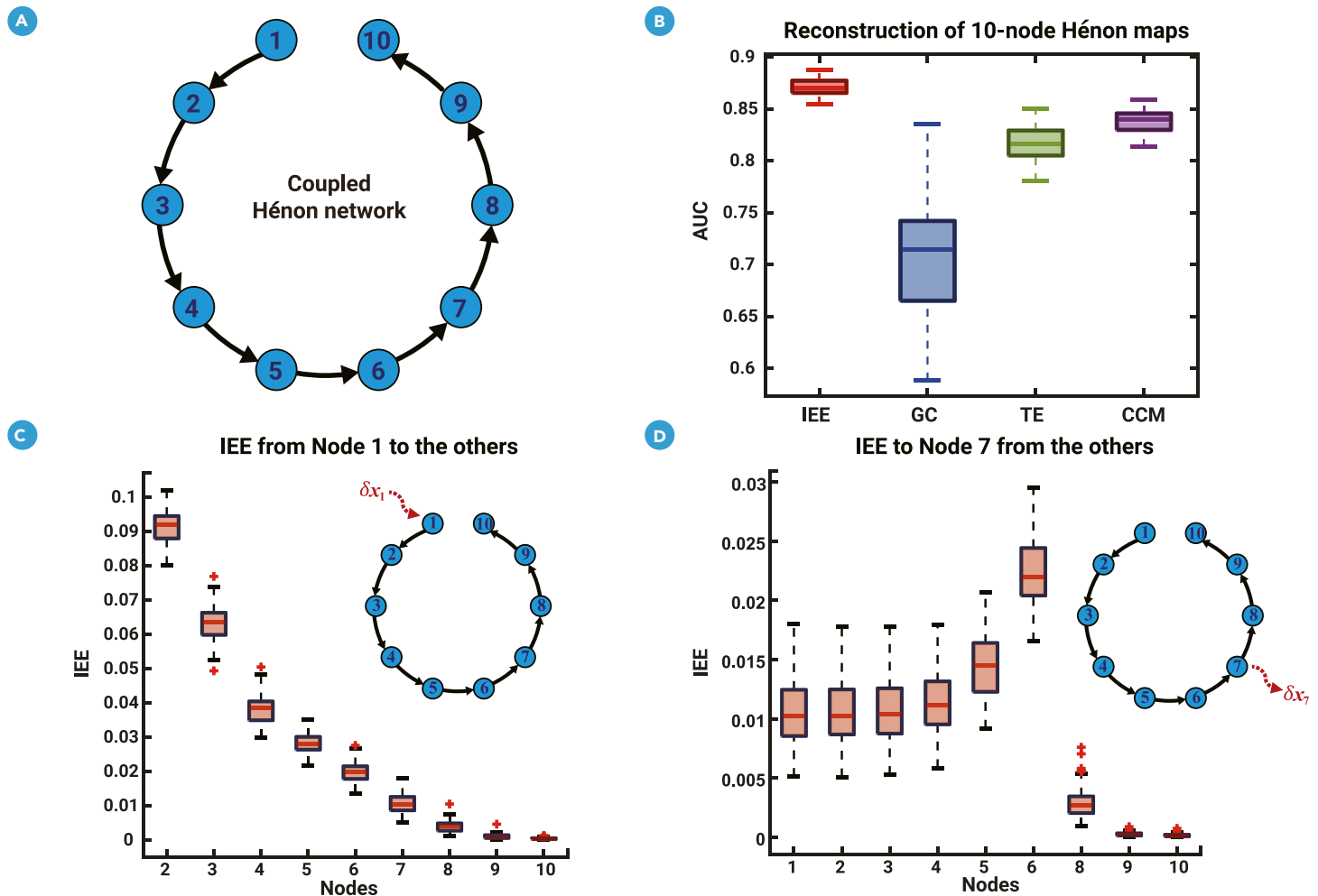


Figure 3. Performance of IEE on the 10-node coupled-Hénon-map dynamics (A) The network structure of the dynamics. There is directed causality from each node to its subsequent node. (B) A boxplot of the area under the curve (AUC) values for the network reconstruction. IEE (0.871 ± 0.008) outperforms GC (0.707 ± 0.069), TE (0.816 ± 0.018), and CCM (0.837 ± 0.012) significantly in accuracy and stability. (C and D) (C) The IEE from node 1 to the other nine nodes. (D) The IEE received by node 7 from the other nodes. IEE accurately captures the interventional information flow and IntDC in the cascade topological network.

($\rho_{\tau}^{\text{IEE}} = 0.383$), GC ($\rho_{\tau}^{\text{GC}} = 0.269$), and TE ($\rho_{\tau}^{\text{TE}} = 0.379$). These results indicate that IEE can accurately quantify IntDC from observational data $\mathbf{x}_{\text{obs}}$ alone and reflect KLD in a linear manner, alleviating the need for additional interventions to the dynamics.

Application of IEE in inferring neural connectomes of *C. elegans*

We applied the IEE criterion to infer the neural connectomes of *C. elegans*, a model organism known for its comprehensively studied nervous system.⁴⁹ Calcium fluorescence imaging time-series data from 31 neurons with specific functions of a freely moving *C. elegans* were collected from Kato et al.⁵⁰ (see Figures 5A, 5B, and S7). The cosine similarities between different neurons were calculated and the neurons were clustered into seven clusters after data preprocessing (see Figure 5C, supplemental information, and Table S5). We presented the ground truth of neural connectomes between clusters in a directed graph (Figure 5D), which was detected by electron microscopy.^{49,51,52} The IEE values between different clusters were calculated to infer the IntDC network. The ROC curve of IEE got a high AUC value of 0.882 (Figure 5F). By maximum Youden index, maximum concordance probability, and minimum distance to the point (0,1), IEE gave the same optimal operating point (OOP). At the OOP, IEE provided an inferred connectomic network with only one false positive and four false negatives (Figure 5E). We also compared IEE with other ConDC indices, i.e., GC/TE/CCM, which revealed that IEE exhibited the highest AUC value, best OOP, largest similarity to the ground truth, lowest false positive at the OOP, and lowest false negative at the OOP (see Table 2; Figures S8–S10; Table S6). Unidirectional edges are particularly sensitive to threshold selection and are prone to false-positive and false-negative inference.

In the *C. elegans* network, there are six unidirectional edges among the 26 directed edges (edges $2 \rightarrow 3$, $2 \rightarrow 4$, $2 \rightarrow 5$, $7 \rightarrow 4$, $7 \rightarrow 5$, $7 \rightarrow 6$ in Figure 5D). CCM and IEE showed comparable performance for bidirectional edges, while IEE achieves the lowest error rate for unidirectional edges compared with other indices (Figures 5 and S10; Table S6).

Additionally, we inferred the connectomic network using the PCMC algorithm with two approaches: the partial correlation (PCMC-ParCorr) and CMI test (PCMC-CMI). PCMC used an independence test with the only parameter being the significance level α_{PC} . The false-positive edges for PCMC-ParCorr ($\alpha_{\text{PC}} = 0.05$) and PCMC-CMI ($\alpha_{\text{PC}} = 0.01$) are 5 and 6, respectively. Both approaches have four false-negative edges. A detailed description of the procedure and results can be found in the supplemental information, Figures S10E and S10F, and Table S6.

These findings demonstrate the efficacy of IEE in reconstructing the neural connectomes of *C. elegans*.

Application of IEE to COVID-19 transmission in Japan

We validated that IEE was a promising indicator for assessing the transmission dynamics of infectious diseases. Our study collected the daily confirmed COVID-19 cases from all 47 prefectures in Japan, spanning a duration of 1,209 days from January 16, 2020 to May 8, 2023 (Figure S11).

First, we calculated the IEE indices for IntDC among different prefectures. IEE effectively ranked the influence across prefectures, with the top five affected areas by Tokyo being Kanagawa (with IEE value 0.601), Chiba (with IEE value 0.568), Saitama (with IEE value 0.534), Aichi (with IEE value 0.453), and Osaka (with IEE value 0.435). Remarkably, this ranking coincides with the geographical

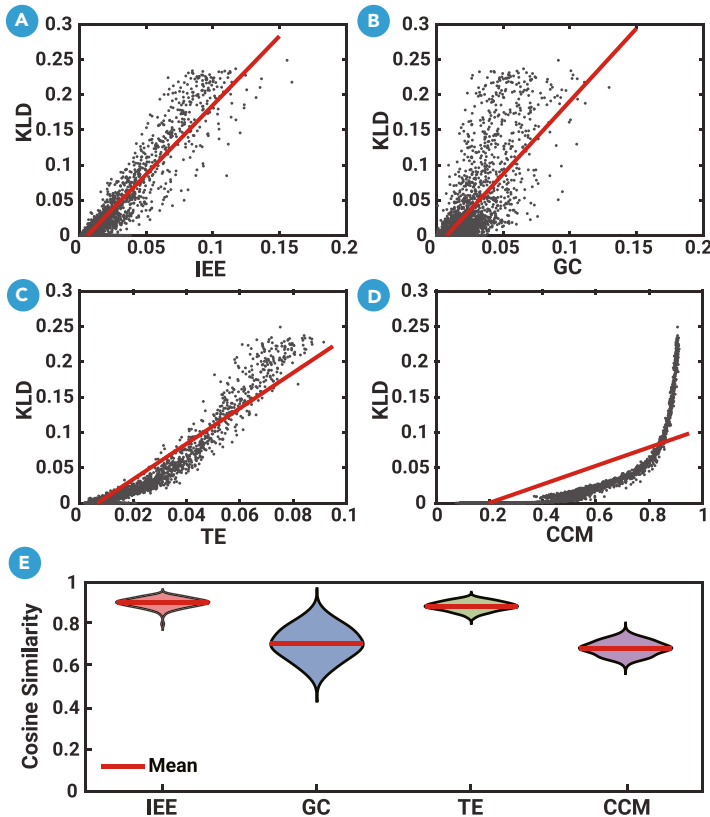


Figure 4. The performance of IEE when measuring the influence of perturbations on the chNNs The Kullback-Leibler divergence (KLD) is utilized to measure the true influence or IntDC between neurons. In (A)–(D), 1,000 samples for different perturbations on chNNs are represented by gray dots, along with the linear regressions (the red lines) between KLD and IEE/GC/TE/CCM, respectively. (E) Violin plots of cosine similarity between KLD and IEE/GC/TE/CCM. The mean values and standard deviations of the cosine similarity are IEE (0.904 ± 0.024), GC (0.711 ± 0.076), TE (0.888 ± 0.022), and CCM (0.687 ± 0.033). IEE can linearly reflect the KLD with high similarity, indicating its effectiveness in accurately quantifying IntDC without additional perturbed data on chNNs.

proximity and socio-economic connections to Tokyo. Kanagawa, Chiba, and Saitama are in the same metropolitan area with Tokyo, while Aichi and Osaka are far from Tokyo but connected to Tokyo by expressway and Shinkansen.

Then, IEE was validated to coincide with the effective distance matrix D^{COVID} between prefectures. The $D^{\text{COVID}} \in \mathbb{R}^{47 \times 47}$ was non-symmetric and designed by incorporating factors such as geometric distances, human mobility, population sizes, and infectious rates across prefectures (based on the gravity model^{53,54}; see supplemental information and Figure S11 for details). The element D_{ij}^{COVID} served as a benchmark for quantifying COVID-19 transmission from prefecture i to j . The IEE values demonstrated a strong linear correlation with $\text{InD}^{\text{COVID}}$, particularly when assessing the causality from Tokyo/Osaka to other prefectures (with Pearson correlation coefficients, i.e. PCCs, -0.906 for Tokyo and -0.860 for Osaka; Figures 6A and 6B). This correlation remained high for other regions (see Figure S12), such as Aichi (with PCC value -0.811), Hokkaido (with PCC value -0.751), Fukuoka (with PCC value -0.752), and Okinawa (with PCC value -0.811).

Further, IEE outperformed ConDC indices in practicality. Notably, the absolute value of the PCC between IEE and $\text{InD}^{\text{COVID}}$ was 0.724 substantially higher than those obtained by GC 0.161, TE 0.284, and CCM 0.568 (average over specific prefecture; see the second to last column in Table 3 and the boxplot in Figure 6C). More detailed comparisons to support the effectiveness of IEE are presented in Table 3 (with specific illustration for Tokyo in Figure S13).

IEE can also reflect asymmetry causal influence (i.e., effective distance) with high PCC under the IntDC framework, highlighting its advantage over ConDC methods. For example, the effective distance or IEE from Tokyo to Osaka ranks fifth (Figure 6A), whereas the reverse direction from Osaka to Tokyo ranks second (Figure 6B).

These results underscore the reliability and efficacy of IEE in providing quantitative insights into COVID-19 transmission dynamics in Japan. Importantly,

IEE performs as a simple and valuable tool for causal analyses solely from daily confirmed time-series data, without the need for complex models considering various factors such as geodesic distances, human mobility, population sizes, and infectious rates.

Application of IEE to investigating circadian rhythms

We investigated the gene regulatory networks (GRNs) related to key genes on the circadian rhythm. The time series of gene expressions were measured by microarray from cultured rat cells.^{55,56} Through decades of molecular and genetic studies, many key circadian genes, such as *Clock*, *Bmal1*(*Arntl*), *Dec1*(*Bhlhb2*), *Dec2*(*Bhlhb3*), *Cry1*, *Cry2*, *Per1*, *Per2*, *Per3*, have been identified in mammals.^{57,58} Figure 6D displays the GRN around *Clock* at the protein level, where the transcription factor *Clock* is phosphorylated by PFK family genes. We calculated the IEE/GC/TE/CCM to reconstruct the GRN (ROC curves and AUC values in Figure 6E). IEE designed for quantifying IntDC presented a higher AUC value (0.737) than ConDC indices (0.490 for GC, 0.610 for TE, and 0.644 for CCM). We also tested the indices on the GRN surrounding *Cry1/Cry2* (Figure S14), where IEE obtained the highest AUC value (0.639) and outperformed the others (0.428 for GC, 0.619 for TE, and 0.614 for CCM).

CONCLUDING REMARKS AND PERSPECTIVE

In summary, we have established a framework of IntDC based on dynamical systems theory and introduced the IEE criterion to quantify IntDC in this study. IEE measures the information flow between interventional causes and effects in the delay-embedding space, making it suitable for analyzing nonlinear and non-separable systems.^{10,14,32} Moreover, IEE is able to infer IntDC solely from observational time-series data, without requiring additional perturbations to the system. This property makes IEE particularly suitable for non-intervention systems. Both theoretical derivations and numerical examples presented in this study provide strong evidence to support the effectiveness, accuracy, and robustness of IEE in detecting IntDC and reconstructing networks. Furthermore, IEE serves as an effective tool for evaluating and ranking the causal dependence between variables within a dynamical system. Through real-world examples, we illustrate the promising applicability of IEE in diverse fields such as regulatory inference and disease-transmission studies.

Information-flow-based methods have been developing rapidly in recent years.^{33,34,36} Liang's information-flow theory, albeit with full nonlinearity in its theoretical formalism, is yet to be implemented for practical applications on real-world data. Existing implementations still rely on assumptions such as independent white noise and linearity when evaluating information flow from data. Our IEE algorithm, inspired by the conception information flow, is specifically designed to handle fully nonlinear systems and has demonstrated applicability across a variety of real datasets. The proposed IntDC framework, together with the IEE algorithm, advances the frontier of causality research by providing a practical and effective tool for nonlinear causal inference.

Runge et al.⁴⁶ summarized recent advances and key challenges in the field of causal inference and developed the platform <https://causeme.net>, which serves as a valuable resource for researchers. Causal inference holds great promise for applications in Earth system science and beyond. Our proposed IEE algorithm, together with the concept of IntDC, provides a novel and practical approach for investigating causality in complex nonlinear systems.

There are several important open issues that warrant further investigation and future development.

- (1) The theory of IntDC and the IEE criterion can be extended to analyze causal relations between groups of variables. Specifically, in Equation 2, the variables x and y can be generalized to multivariate vectors, enabling the investigation of complex interactions among multiple variables.
- (2) The data used for computing IEE can be non-uniformly sampled in time. In this study, we employed the standard time-delayed embedding Equation 3 and Equation 4. However, according to the generalized Takens' embedding theorem, any set of $L + 1$ observations of x_t and y_t can construct topological diffeomorphisms to the manifolds $\mathcal{M}_x$ and $\mathcal{M}_y$, respectively. As shown in Equation 7, only the local neighborhood structure in the embedding space is essential for the computation. Therefore, the IEE algorithm can be naturally

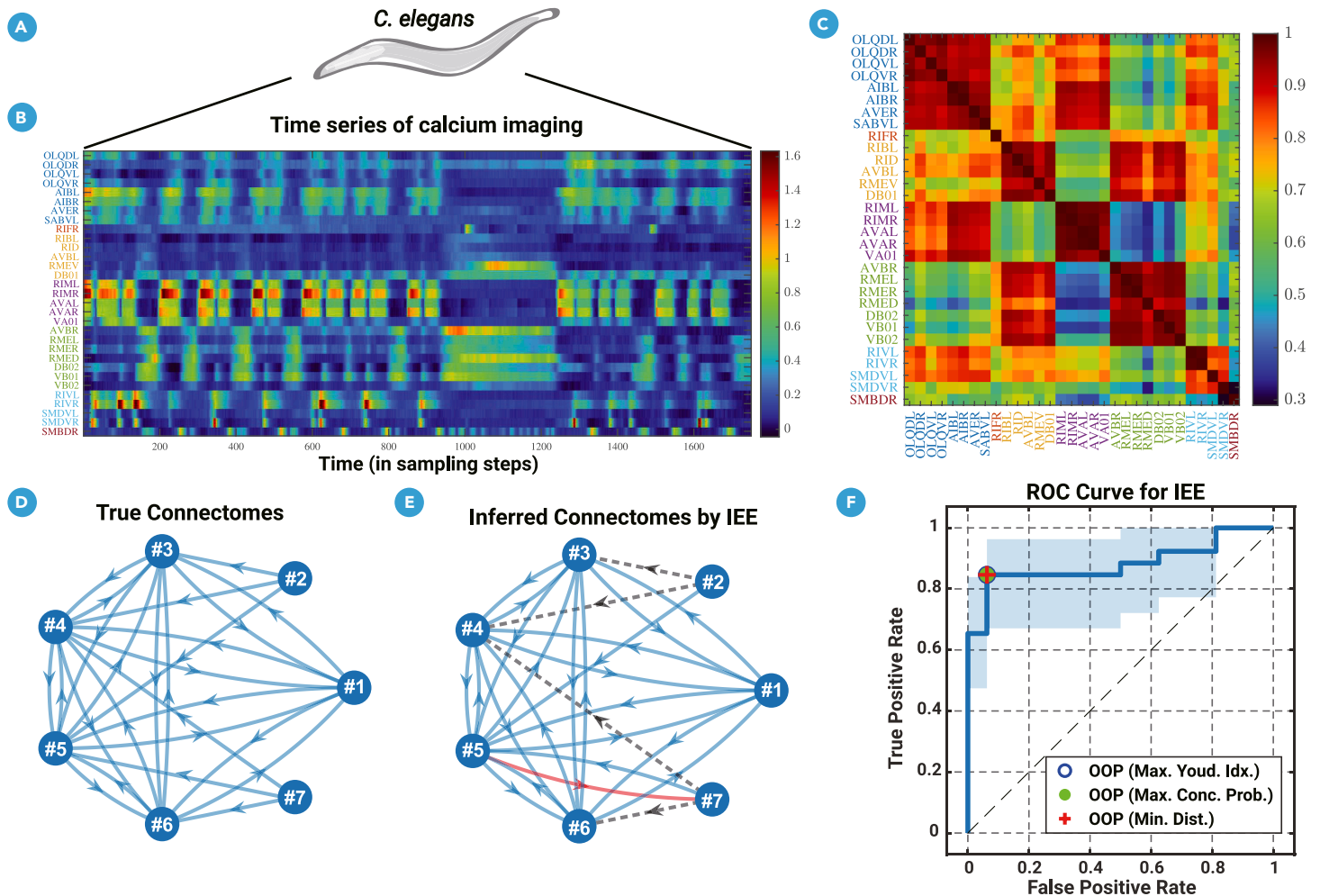


Figure 5. Application of the IEE on inferring the neural connectomes of *C. elegans* (A) Illustration of *C. elegans*. (B) The calcium imaging time series of 31 key neurons, with their names listed on the left and colors representing different clusters. (C) The cosine similarity matrix between neurons, aiding the clustering process. (D and E) (D) The true connectomes between seven clusters. (E) The inferred causal network at the optimal operating point (OOP) determined by IEE. The red edge represents a false positive, and the black dashed edges represent false negatives. (F) The ROC curve (the blue line) for IEE. The same OOP is obtained under three criteria, i.e., the maximum Youden index (the blue circle), the maximum concordance probability (the green dot), and the minimum distance to the point (0,1) (the red cross). The shaded area around the ROC curve represents the 95% confidence interval obtained through bootstrapping. The AUC value of IEE is 0.882.

extended to infer causality from datasets sampled at varying or irregular time intervals.

- (3) Although interventions in the admissible state space are allowed in Theorem 2, perturbations in the tangent space are sufficient to estimate IntDC. Under the stationarity assumption (A5), all trajectories with admissible initial points converge to the attractive manifold, and data on the attractor together with tangent-space perturbations are sufficient to quantify Equation 6.
- (4) To further distinguish direct and indirect causality, we can generalize IEE to its conditional version

$$\text{cIEE}[x \rightarrow y|z] := \text{CMI}(\delta \mathbf{X}_t, \delta \mathbf{Y}_{t+1} | \mathbf{Y}_{t+1}, \mathbf{Z}_t, \delta \mathbf{Z}_t), \quad (\text{Equation 11})$$

where “cIEE” is short for the conditional IEE, z is a third variable whose time-delayed vector is $\mathbf{Z}_t$, and $\delta \mathbf{Z}_t$ represents the intervened deviation on $\mathbf{Z}_t$. Equation 11 measures the direct IntDC from x to y conditioned on z . Further, in combination with the well-known PC algorithm⁵⁹ or PCMC, cIEE offers a feasible way to remove high-order indirect causal edges iteratively and reconstruct the direct causal network.

- (5) Researchers have pointed out that time-delayed embedding algorithms are particularly suitable for deterministic dynamical systems with an attractive manifold.^{14,17,26} However, the coupling strength between variables could influence the accuracy of these methods. Stark et al.³⁹ extended Takens' embedding to stochastic systems, providing a theoretical basis for applying embedding approaches beyond purely

deterministic dynamics. In the results and discussions section of this study, we demonstrate the effectiveness of IEE for logistic systems with Gaussian noises. Nevertheless, how systematic factors, such as coupling strength and noise characteristics, affect the quantification of causal strength in more general settings remains an open question.

- (6) We provide a concise introduction to the field of dynamical causality. The dynamical causality framework¹⁰ summarized causal inference based on dynamical systems. Methods such as GC,¹¹ TE,¹³ CCM,¹⁴ and embedding entropy¹⁰ are typical examples of dynamical causality approaches. Dynamical causality, or ConDC in this study, incorporates temporal ordering, allows feedback and causal loops, and quantifies causal strength—representing an advance over statistical causal inference. IntDC in this study further extended this framework by introducing interventions, and IEE provides an efficient algorithm for quantifying IntDC in non-intervention complex systems.
- (7) One limitation of IEE is its requirement for long-time-series data, typically over 100 time points. Some datasets, such as spatial transcriptomes,⁶⁰ contain many samples along the spatial dimension but few time points. A potential approach to address this limitation is to assign pseudo-time labels to these samples.^{61,62} By converting the data into a time-series format, it may become possible to infer causal relationships between variables. This represents an important direction for future work.

Table 2. Comparison of causal indices on the inference of *C. elegans* neural connectomes

	Properties of ROC curves				Cosine similarity	
	AUC	Maximum Youden index	Maximum concordance probability	Minimum distance to (0,1)	To <i>C</i>	To log(1 + <i>C</i>)
C^{IEE}	0.882	0.784	0.793	0.166	0.750	0.905
C^{GC}	0.858	0.721	0.740	0.198	0.430	0.678
C^{TE}	0.796	0.620	0.656	0.269	0.477	0.726
C^{CCM}	0.880	0.615	0.639	0.294	0.644	0.866

The matrices $C^{IEE}/C^{GC}/C^{TE}/C^{CCM}$ represent the inferred connectomes by the four indices. The matrix *C* is the ground truth of neural connectomes. The best value in each column is shown in bold.

(8) Moreover, causal inference empowered by deep learning is emerging as a frontier in research. Notable advances include CausalEGM,⁶³ Causalformer,⁶⁴ intervened reservoir computing,⁶⁵ and reservoir cross-mapping.⁶⁶ In the authors' view, AI and logistic-based causal inference methods have distinct advantages. Logistic-based methods are more interpretable and provide mechanistic insights, while AI methods are highly efficient at discovering complex correlations or mappings between different parts of a complex system. However, these approaches are also complementary: combining AI with logistic-based method should provide a more comprehensive understanding of complex systems. Integrating the IntDC framework with neural networks presents a promising direction for further exploration. On one hand, causality theory provides interpretability to AI; on the other hand, large AI foundation models offer unprecedented capacity to uncover causal relationships in complex data. These developments signal the advent of a transformative "big causality era," in which data-driven discovery and causal reasoning evolve in synergy.

Data and code availability

• The code for IEE is available at <https://github.com/smsxiaomayi/IEE>.

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AUTHOR CONTRIBUTIONS

J.S. contributed to the conceptualization, methodology, investigation, computation, data analysis, visualization, and writing of the original draft. Y.L. contributed to the investigation, computation, data analysis, visualization, and writing of the original draft. J.Z. contributed to the investigation, computation, data analysis, and visualization, and writing of the original draft. S.L. contributed to the investigation, computation, data analysis, and visualization. R.B. contributed to the investigation and data analysis. K.A. contributed to the conceptualization, investigation, and supervision. L.C. contributed to the conceptualization, investigation, and supervision. W.L. contributed to the conceptualization, investigation, and supervision. All authors contributed to the manuscript and approved the final version.

DECLARATION OF INTERESTS

The authors declare no competing interests.

SUPPLEMENTAL INFORMATION

It can be found online at <https://doi.org/10.1016/j.xinn.2026.101358>.

RESOURCE AVAILABILITY

Materials availability

No material was generated in the present study.

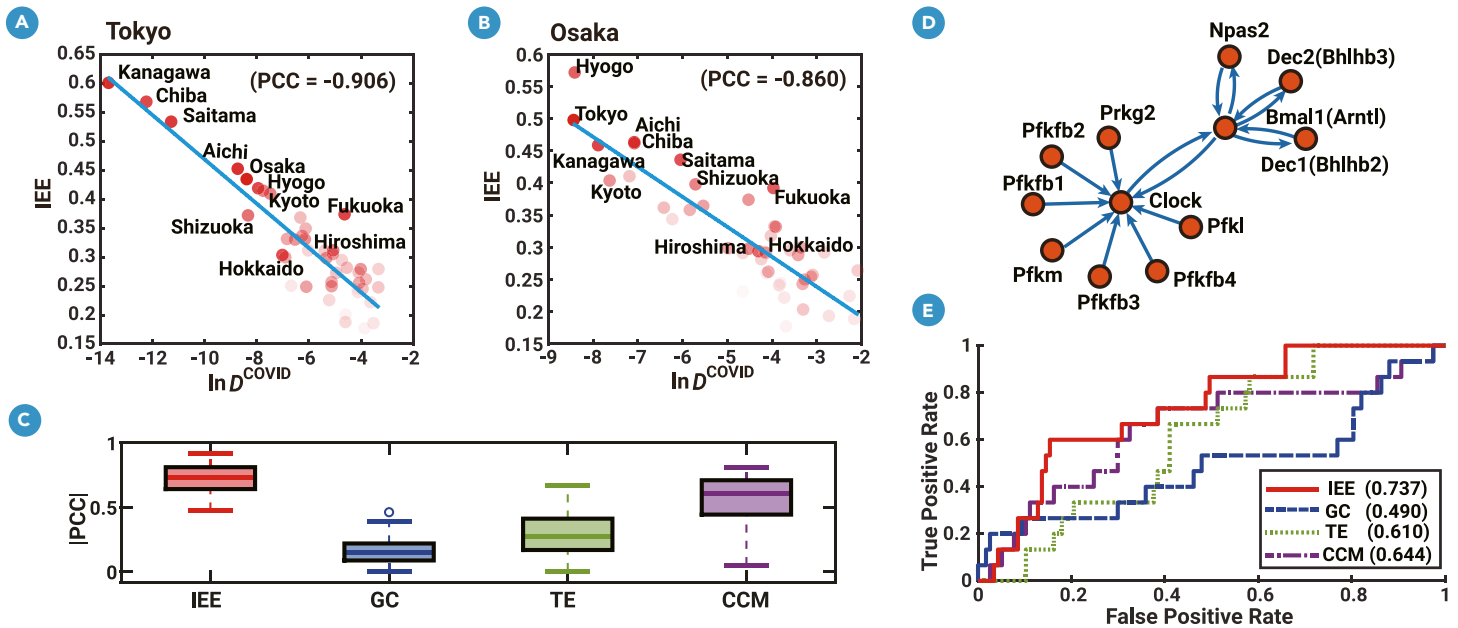


Figure 6. Application of the IEE on the COVID-19 transmission in Japan and the GRN of circadian rhythms (A) IEE values from Tokyo to other prefectures are highly linearly correlated with $\ln D^{COVID}$, where D^{COVID} is the effective distance obtained from the gravity model in consideration of various factors. (B) The result for IEE values from Osaka to the other prefectures. (C) The boxplot for $|PCC(c_i^{inf}, \ln D_i^{COVID})|$ with average over fixed *i*, where PCC is the Pearson correlation coefficient and c_i^{inf} is the inferred causal strengths by IEE, GC, TE, or CCM. The PCC linearity of IEE (0.724 ± 0.116) is significantly higher than that of GC (0.161 ± 0.099), TE (0.284 ± 0.168), and CCM (0.568 ± 0.174). (D) The GRN surrounding *Clock*, a key circadian gene. (E) The ROC curves and AUC values of the IntDC index IEE, and three ConDC indices GC/TE/CCM.

Table 3. Comparison of IEE/GC/TE/CCM in measuring causal strengths in the transmission of COVID-19 data in Japan

	$\left PCC\left(c_{ij}^{inf}, \ln D_{ij}^{COVID}\right)\right $ for specific prefecture (fixed i)							$\left PCC\left(c_{ij}^{inf}, \ln D_{ij}^{COVID}\right)\right $ For all i, j with $i \neq j$
	Tokyo ($i = 13$)	Osaka ($i = 27$)	Aichi ($i = 23$)	Hokkaido ($i = 1$)	Fukuoka ($i = 40$)	Okinawa ($i = 47$)	Average Over all i	
IEE	0.906	0.860	0.811	0.751	0.752	0.811	0.724	0.748
GC	0.462	0.387	0.236	0.070	0.125	0.164	0.161	0.166
TE	0.655	0.308	0.018	0.445	0.526	0.538	0.284	0.109
CCM	0.799	0.795	0.448	0.052	0.727	0.477	0.568	0.585

i and j (ranging from 1 to 47) stand for different prefectures in Japan. c_{ij}^{inf} is the causal strength from the prefecture i to j inferred by IEE, GC, TE, or CCM. D_{ij}^{COVID} is the effective distance between prefectures i and j . PCC is the Pearson correlation coefficient. The best value in each column is shown in bold.

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